# Chapter 1: Comparison of Batches

Applied Multivariate Statistical Analysis

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# Introduction

# What Is Multivariate Statistical Analysis?

**Setting:** We observe $n$ data points in $\mathbb{R}^p$:

$$x_i = (x_{i1}, x_{i2}, \dots, x_{ip}), \quad i = 1, \dots, n,$$

each a realisation of a random vector $X = (X_1, X_2, \dots, X_p)^\top \in \mathbb{R}^p$.

**Key questions addressed by descriptive techniques:**

- Are some components more spread than others?
- Do sub-groups of observations exist?
- Are there outliers?
- How "normal" is the distribution?
- Are there low-dimensional linear combinations of $X$ with non-normal behaviour?

# Perceptual Challenges in High Dimensions

- **2D:** Point clouds are immediately interpretable.
- **3D:** Real-time rotation on a computer screen is feasible.
- **4D:** A “sliding” technique (Härdle & Scott, 1992) can show 3D density contours as the fourth variable varies.
- $\geq 5$**D:** A qualitative jump in difficulty occurs; high-dimensional structure must be mapped into lower dimensions (Klinke & Polzehl, 1995).

**This chapter** introduces:

- Boxplots
- Histograms
- Kernel density estimates
- Scatterplots & draftman plots
- Chernoff–Flury faces
- Andrews’ curves
- Parallel coordinates plots (PCP)

# The Swiss Bank Note Dataset

**Swiss bank notes** (Flury & Riedwyl, 1988): $n = 200$ measurements — 100 **genuine**, 100 **counterfeit**.

Six measurements per note:

$$X_1 = \text{length of bill}$$

$$X_2 = \text{height (left)}$$

$$X_3 = \text{height (right)}$$

$$X_4 = \text{inner frame to lower border}$$

$$X_5 = \text{inner frame to upper border}$$

$$X_6 = \text{diagonal of central picture}$$

**Goal:** Use these measurements to determine whether a bill is genuine or counterfeit.

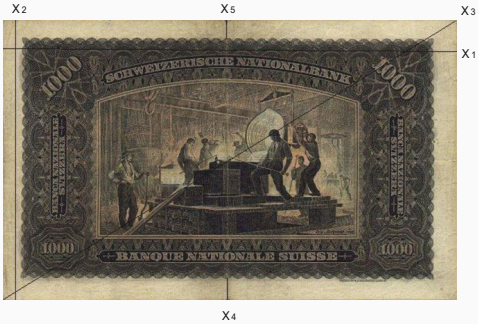



**Figure 1:** Swiss 1000-franc bank note with measurement locations

# Section 1: Boxplots

# The Five Number Summary

**Example:** 15 largest world cities in 2006 (population in 10,000s).

| City | Country | Pop. (10k) | Order stat. |
|---|---|---|---|
| Tokyo | Japan | 3,420 | $x_{(15)}$ |
| Mexico City | Mexico | 2,280 | $x_{(14)}$ |
| Seoul | S. Korea | 2,230 | $x_{(13)}$ |
| New York | USA | 2,190 | $x_{(12)}$ |
| Sao Paulo | Brazil | 2,020 | $x_{(11)}$ |
| Bombay | India | 1,985 | $x_{(10)}$ |
| Delhi | India | 1,970 | $x_{(9)}$ |
| Shanghai | China | 1,815 | $x_{(8)} \leftarrow M$ |
| $\vdots$ | | $\vdots$ | $\vdots$ |
| Karachi | Pakistan | 1,430 | $x_{(1)}$ |

**Median** ($n = 15$ odd): $M = x_{(8)} = 1{,}815$.

# Quartiles and $F$-Spread

**Depth of the fourth:**

$$\text{depth of fourth} = \frac{[\text{depth of median}] + 1}{2} = \frac{8+1}{2} = 4.5$$

So the **fourths** (quartiles) are:

$$F_L = \tfrac{1}{2}\{x_{(4)} + x_{(5)}\} = 1{,}610, \qquad F_U = \tfrac{1}{2}\{x_{(11)} + x_{(12)}\} = 2{,}105.$$

$F$-**spread** and **outside bars** (outlier thresholds):

$$d_F = F_U - F_L = 495$$
$$F_L - 1.5\, d_F = 1{,}610 - 742.5 = 867.5$$
$$F_U + 1.5\, d_F = 2{,}105 + 742.5 = 2{,}847.5$$

$\Rightarrow$ **Tokyo** (3,420) is beyond the upper outside bar $\Rightarrow$ **outlier**.

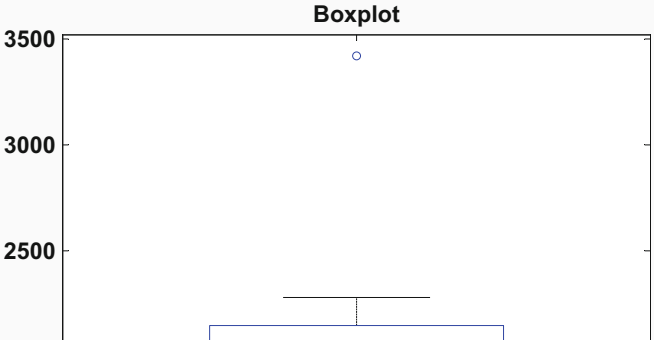


**Five Number Summary:** $\{1{,}430;\ 1{,}610;\ 1{,}815;\ 2{,}105;\ 3{,}420\}$ Mean: $\bar{x} = 1{,}939.7$

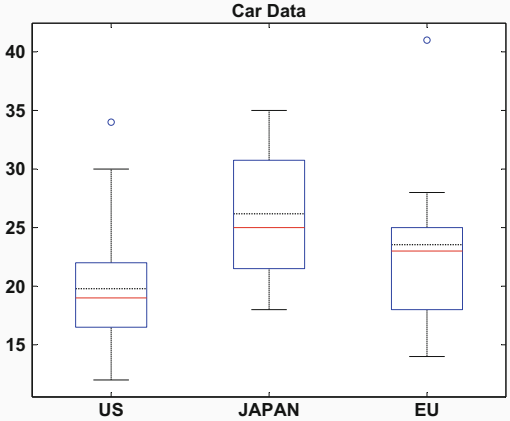
# Constructing the Boxplot

Algorithm:

1. Draw a **box** with borders at $F_L$ and $F_U$ (contains 50% of the data).
2. Draw the **median** as a solid line | and the **mean** as a dotted line ⋮.
3. Draw **whiskers** from each box end to the most remote point that is *not* an outlier.
4. Mark outliers as ⋆ (outside $F_{UL} \pm 1.5\, d_F$) or ● (outside $F_{UL} \pm 3\, d_F$).



# Comparing Car Fuel Economies



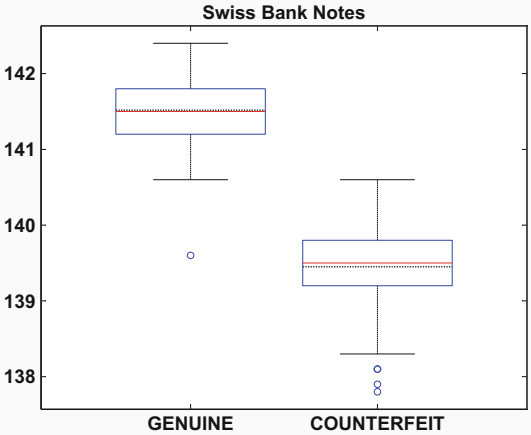

## Five Number Summaries:

| Group | Min | $F_L$ | $M$ | $F_U$ | Max |
|---|---|---|---|---|---|
| US | 12 | 16.8 | 18.8 | 22 | 30 |
| Japan | 18 | 22 | 25 | 30.5 | 35 |
| EU | 14 | 19 | 23 | 25 | 28 |

## Conclusions:

- Japanese cars are most fuel-efficient.
- One EU outlier: VW Rabbit Golf Diesel.
- US box lies entirely below Japanese box.
- Worst Japanese car outperforms $\approx 50\%$ of US cars.

# Swiss Bank Notes: $X_6$ (Diagonal)




Genuine notes ($n = 100$):

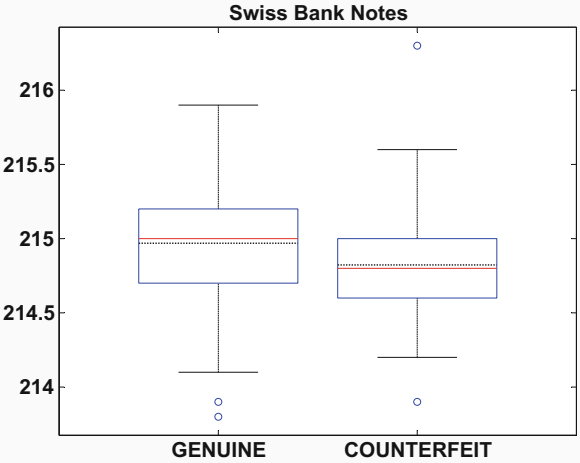
| # | 100 | Genuine | |
|---|---|---|---|
| $M$ | 50.5 | 141.5 | |
| $F$ | 25.75 | 141.25 | 141.8 |
| | 1 | 140.65 | 142.4 |

Counterfeit notes ($n = 100$):

| # | 100 | Counterfeit | |
|---|---|---|---|
| $M$ | 50.5 | 139.5 | |
| $F$ | 25.75 | 139.2 | 139.8 |
| | 1 | 138.3 | 140.65 |

$\Rightarrow$ Genuine diagonals are **systematically larger**, but with some overlap.

# Swiss Bank Notes: $X_1$ (Length)




- Length ($X_1$) shows **less clear separation** than diagonal ($X_6$).
- Distributions largely overlap.
- A few outliers in each group.
- Boxplots alone do **not reveal multimodality**.

# Boxplot Summary

| Summary: Boxplots | |
|---|---|
| ↪ | Median and mean are measures of **location**. |
| ↪ | Position of median/mean within the box measures **skewness**. |
| ↪ | Box length and whisker length measure **spread** and **tail length**. |
| ↪ | Outliers: ⋆ outside $F_{UL} \pm 1.5d_F$; • outside $F_{UL} \pm 3d_F$. |
| ↪ | Boxplots do **not** reveal multimodality or clusters. |
| ↪ | Side-by-side boxplots compare entire **distributions**. |

# Section 2: Histograms

# Definition of the Histogram

**Bin** of length $h$ with origin $x_0$:

$$B_j(x_0, h) = [x_0 + (j-1)h,\ x_0 + jh), \quad j \in \mathbb{Z}.$$

**Histogram estimator** for an i.i.d. sample $\{x_i\}_{i=1}^n$:

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{j \in \mathbb{Z}} \sum_{i=1}^n \mathbf{1}\{x_i \in B_j(x_0, h)\}\, \mathbf{1}\{x \in B_j(x_0, h)\}.$$

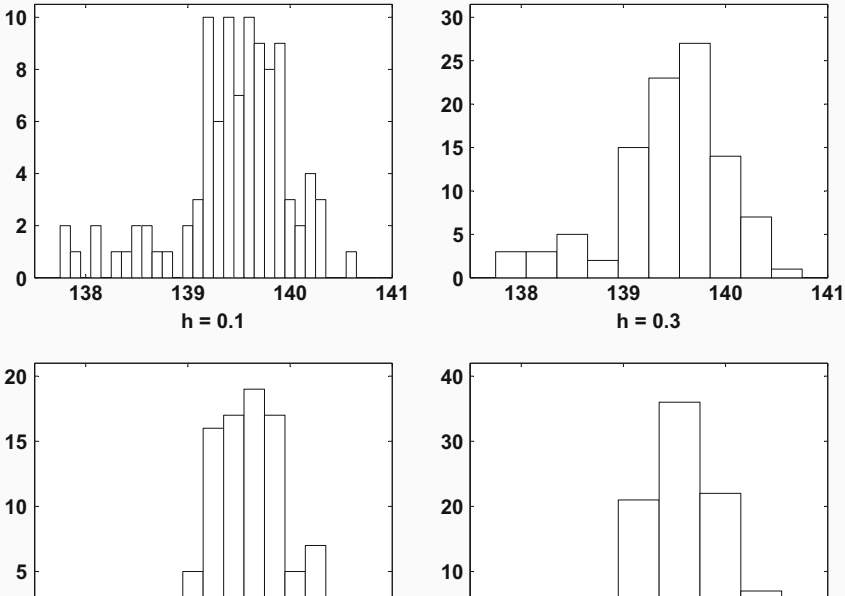
**Key parameter:** binwidth $h$.

- $h$ too **large** $\Rightarrow$ over-smoothed, unstructured histogram.
- $h$ too **small** $\Rightarrow$ noisy histogram with many spurious peaks.

**Optimal binwidth** (rule of thumb):

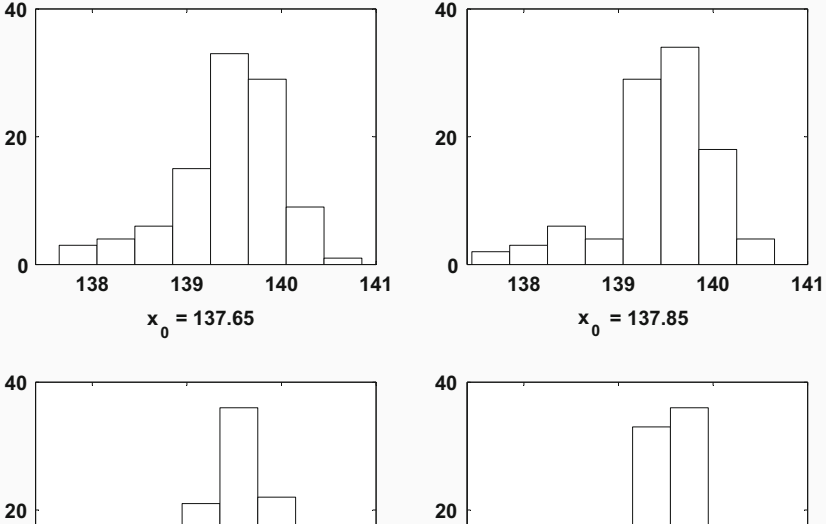
$$h_{\text{opt}} = \left( \frac{24\sqrt{\pi}}{n} \right)^{1/3}.$$

# Effect of Binwidth $h$



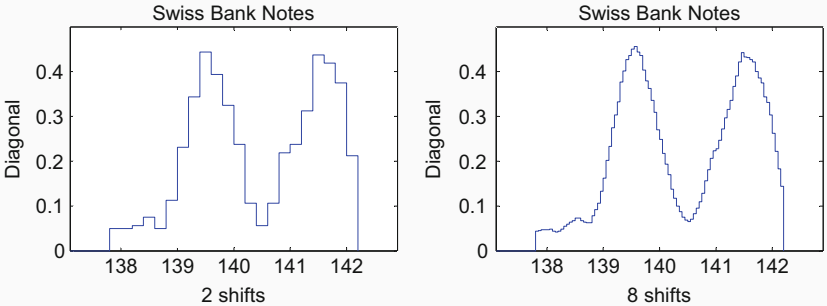
# Effect of Bin Origin $x_0$

**Problem:** Even with fixed $h$, different origins $x_0$ yield different histograms!



# Averaged Shifted Histograms (ASH)

Averaging over multiple histogram origins removes the dependence on $x_0$ and reveals the **bimodal** structure of the diagonal variable.



# Histogram Summary

| Summary: Histograms | |
|---|---|
| $\hookrightarrow$ | Histograms detect **modes**; modes correspond to strong peaks. |
| $\hookrightarrow$ | Same $h$, different $x_0 \Rightarrow$ different histograms. |
| $\hookrightarrow$ | Changing $x_0$ can drastically alter appearance. |
| $\hookrightarrow$ | $h$ too large $\Rightarrow$ flat, unstructured; $h$ too small $\Rightarrow$ unstable. |
| $\hookrightarrow$ | Optimal $h_{\text{opt}} = (24\sqrt{\pi}/n)^{1/3}$ (under normality). |
| $\hookrightarrow$ | Averaged shifted histograms are kernel density estimates. |

# Section 3: Kernel Densities

# From Histogram to Kernel Density

Four critiques of histograms:

1. Binwidth $h$ is hard to choose optimally.
2. Origin $x_0$ also influences the shape.
3. Information is lost (observations replaced by bin centre).
4. The underlying density is smooth, but the histogram is not.

**Key insight:** The histogram at a bin centre can be written as:

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} K\Big(\frac{x - x_i}{h}\Big), \quad K(u) = \mathbf{1}\big(|u| \leq \tfrac{1}{2}\big).$$

Replacing the **uniform kernel** $K$ with a smooth function resolves difficulties (2)–(4).

# Common Kernel Functions

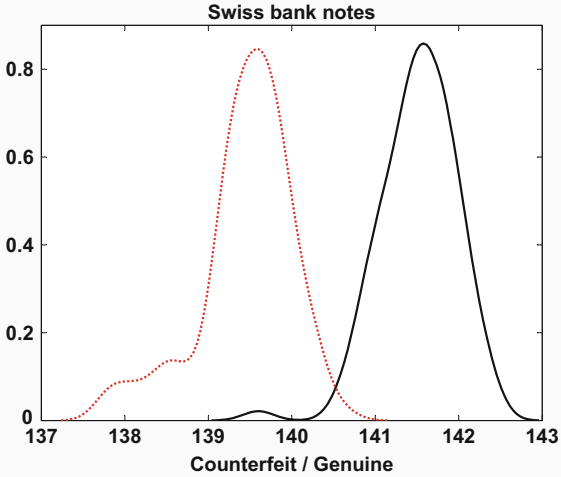
| $K(\bullet)$ | Name |
|---|---|
| $K(u) = \frac{1}{2}\,\mathbf{1}(\lvert u\rvert \le 1)$ | Uniform |
| $K(u) = (1-\lvert u\rvert)\,\mathbf{1}(\lvert u\rvert \le 1)$ | Triangle |
| $K(u) = \frac{3}{4}(1-u^2)\,\mathbf{1}(\lvert u\rvert \le 1)$ | Epanechnikov |
| $K(u) = \frac{15}{16}(1-u^2)^2\,\mathbf{1}(\lvert u\rvert \le 1)$ | Quartic (Biweight) |
| $K(u) = \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{u^2}{2}\right)$ | Gaussian $\varphi$ |

**Bandwidth selection** (Silverman's rule of thumb for Gaussian kernel):

$$h_G = 1.06\,\hat{\sigma}\,n^{-1/5}, \qquad \hat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2}.$$

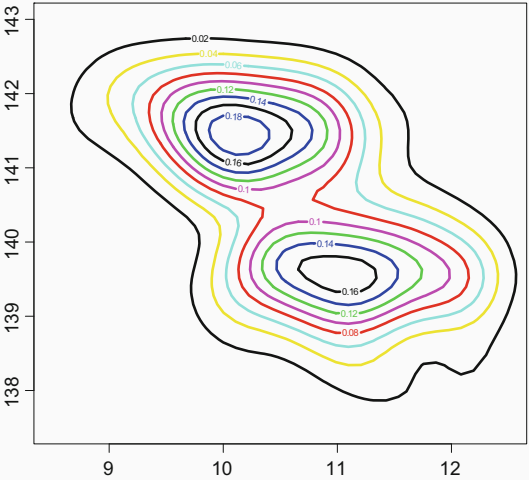
For the quartic kernel: $h_Q = 2.62 \cdot h_G$.

# KDE of Bank Note Diagonals

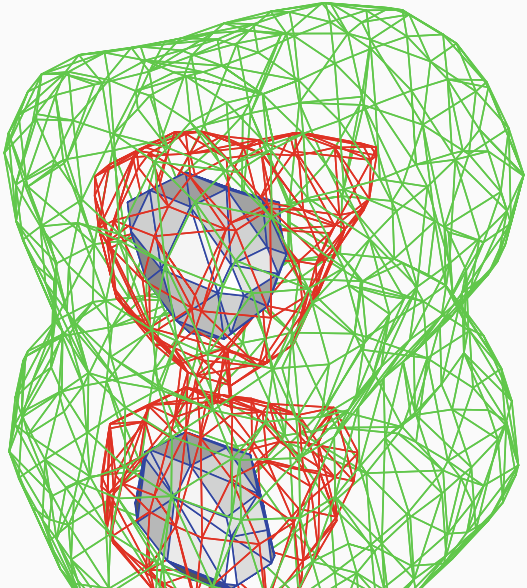



- **Black (solid):** Genuine notes ($X_6 \approx 141.5$).
- **Red (dotted):** Counterfeit notes ($X_6 \approx 139.5$).
- Clear separation, but **overlap** near $\approx 140.4$.
- Cannot perfectly discriminate using $X_6$ alone.
- Need additional variables: $X_4$, $X_5$, ...

# Bivariate and Trivariate Density Estimates




2D: $X_5$ vs $X_6$

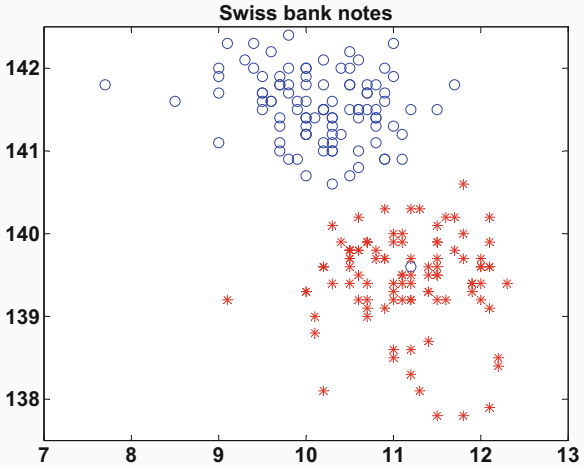


# KDE Summary

| Summary: Kernel Density Estimates | |
|---|---|
| $\hookrightarrow$ | KDEs estimate densities by centring a smooth kernel on each observation. |
| $\hookrightarrow$ | Bandwidth $h$ controls smoothness of $\hat{f}$. |
| $\hookrightarrow$ | KDEs are smooth and can represent distributions up to 3D. |
| $\hookrightarrow$ | Rule of thumb: $h_G = 1.06\hat{\sigma}n^{-1/5}$ (Gaussian kernel only). |
| $\hookrightarrow$ | Good for detecting modes, location, skewness, tails, asymmetry. |

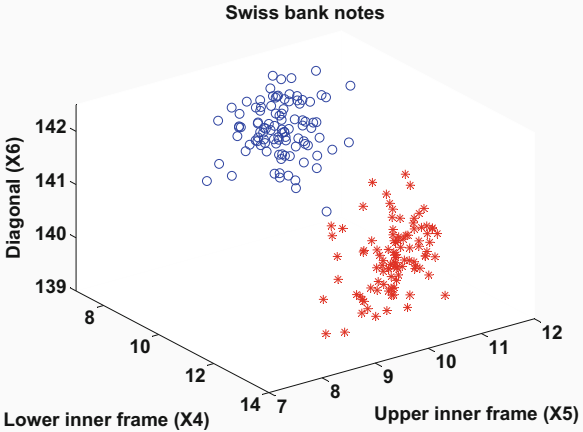
# Section 4: Scatterplots

# 2D Scatterplot: $X_5$ vs $X_6$




- $\circ$ = genuine notes; $*$ = counterfeit notes.
- **Downward-sloping scatter**: larger $X_5$ $\Rightarrow$ smaller $X_6$.
- The **70th observation** (thick circle) is a genuine note embedded in the counterfeit cloud.
- A horizontal separation line would misclassify it.
- Better separation requires **more variables**.

# 3D Scatterplot: $X_4, X_5, X_6$

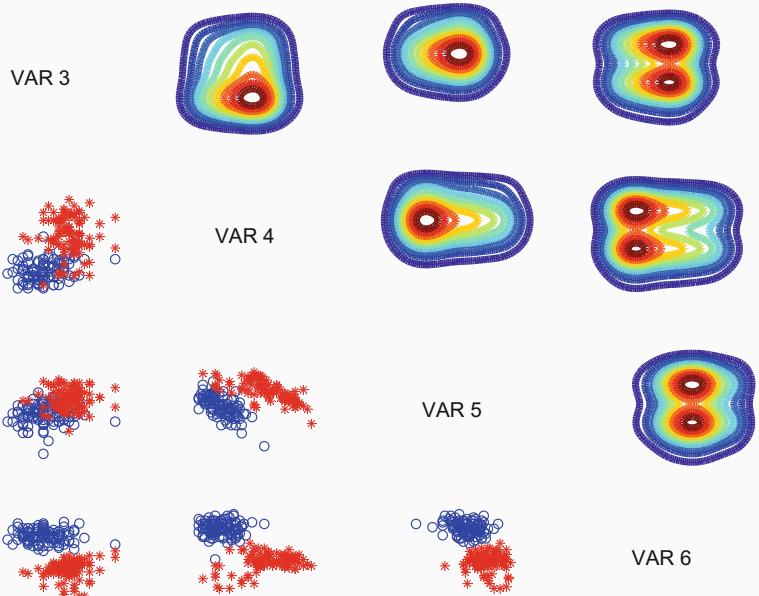



- Adding $X_4$ (lower inner frame distance) gives **better cluster separation**.
- The separation surface is now a **plane**, not a line:

$$a_1 x_1 + \cdots + a_6 x_6 = \text{const.}$$

- The optimal weights $(a_1, \ldots, a_6)$ are found by **discriminant analysis** (Chapter 14).

# Draftman's Plot (Scatterplot Matrix)



# Scatterplot Summary

| Summary: Scatterplots | |
|---|---|
| ↪ | 2D/3D scatterplots identify separated points, outliers, sub-clusters. |
| ↪ | Slope of scatter indicates positive/negative dependence. |
| ↪ | Draftman plot matrices detect structures conditioned on other variables. |
| ↪ | Brushing through a point cloud reveals conditional dependence. |

# Section 5: Chernoff–Flury Faces

# The Idea of Faces

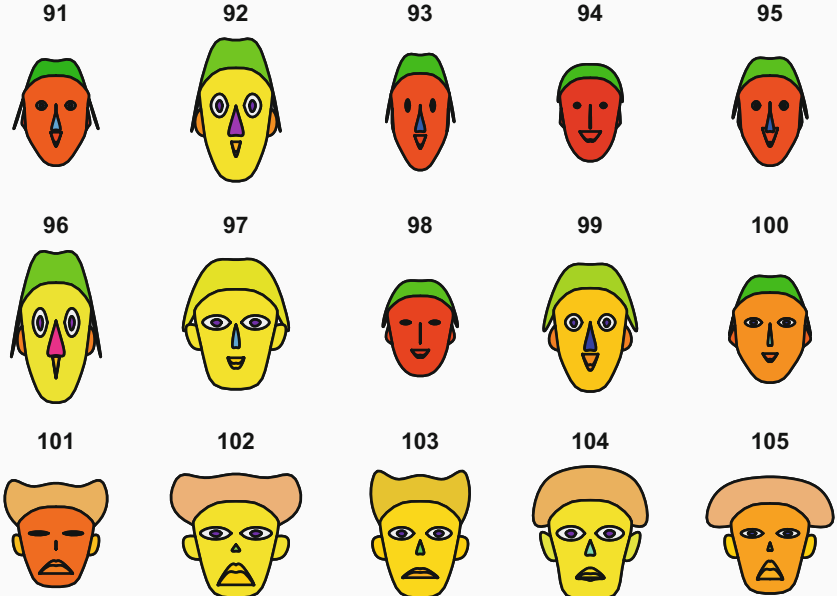
**Principle:** Map each $p$-dimensional observation onto characteristics of a human face, exploiting the brain's powerful face-recognition ability.

- Introduced by **Chernoff (1973)**; extended by **Flury & Riedwyl (1988)**.
- Each variable is rescaled to $[0, 1]$; the two extremes correspond to extreme face element positions.
- Flury–Riedwyl design: **36 face elements** (18 for left side, 18 for right side).

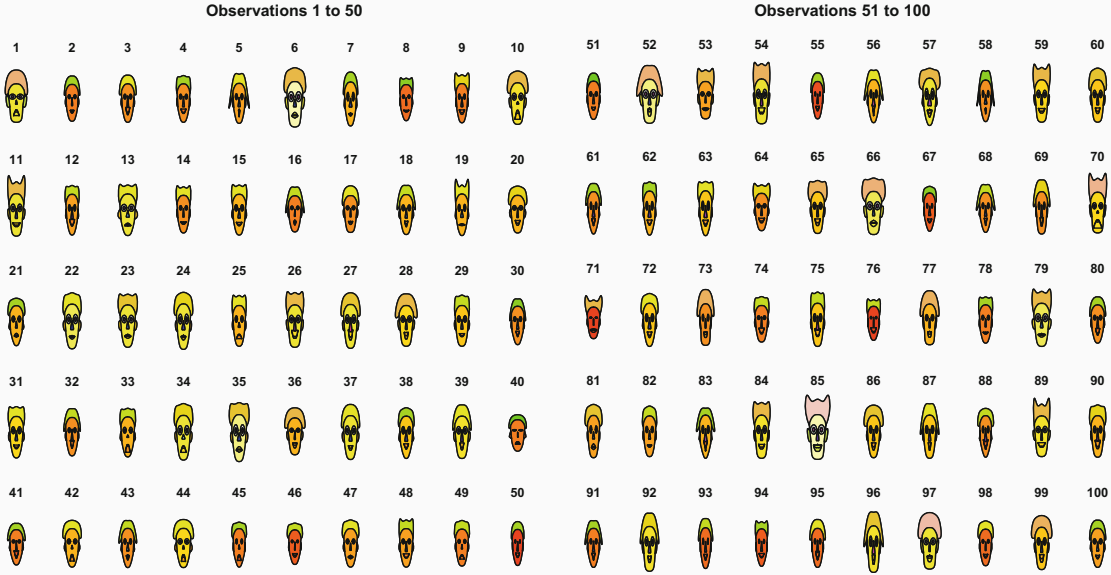
**Bank data assignment** ($p = 6$):

| Variable | Face Element |
|---|---|
| $X_1$ | Eye sizes (1, 19) |
| $X_2$ | Pupil sizes (2, 20) |
| $X_3$ | Eye slants (4, 22) |
| $X_4$ | Upper hair lines (11, 29) |

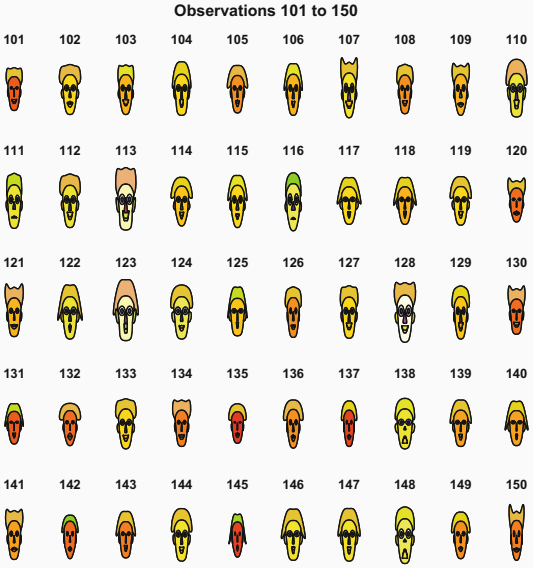
# Faces for Observations 91–110

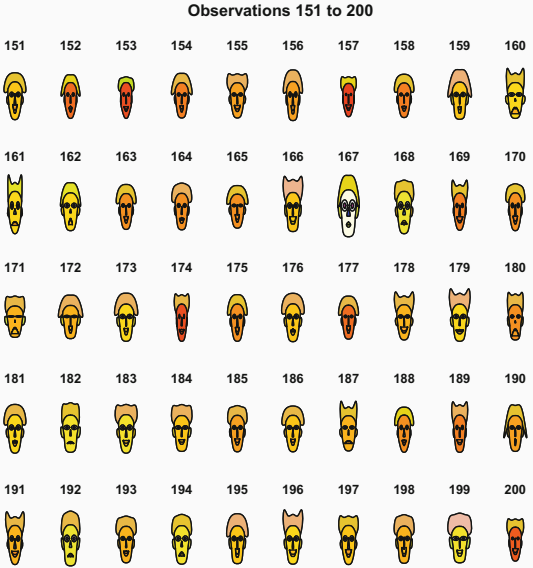


# Faces for All Genuine Notes (Obs. 1–100)



# Faces for All Counterfeit Notes (Obs. 101–200)



# Faces Summary

| Summary: Chernoff–Flury Faces | |
|---|---|
| ↪ | Sub-groups are characterised by **similar-looking faces**. |
| ↪ | Outliers appear as **extreme faces** (e.g. dark hair, big smile). |
| ↪ | If one element of $X$ is unusual, the corresponding face element changes dramatically. |
| ↪ | Variable assignment to face elements matters for interpretability. |

# Section 6: Andrews' Curves

# Definition

**Idea (Andrews, 1972):** Map each $p$-dimensional observation $X_i$ to a Fourier curve:

$$f_i(t) = \begin{cases} \dfrac{X_{i,1}}{\sqrt{2}} + X_{i,2}\sin t + X_{i,3}\cos t + \cdots + X_{i,p}\sin\left(\frac{p}{2}t\right) & p \text{ even} \\ \dfrac{X_{i,1}}{\sqrt{2}} + X_{i,2}\sin t + X_{i,3}\cos t + \cdots + X_{i,p}\cos\left(\frac{p-1}{2}t\right) & p \text{ odd} \end{cases}$$

where $t \in [-\pi, \pi]$.

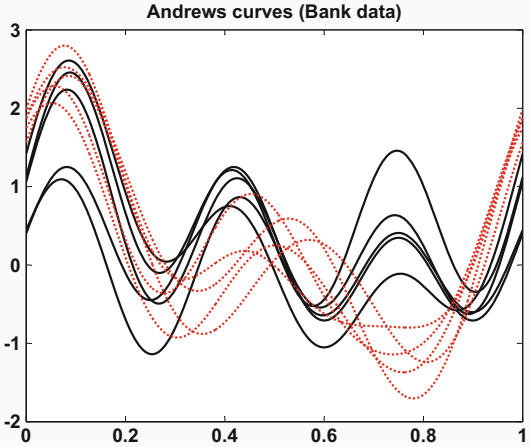
**Example** ($p = 3$): $X_1 = (0, 0, 1)$, $X_2 = (1, 0, 0)$, $X_3 = (0, 1, 0)$ give

$$f_1(t) = \cos t, \quad f_2(t) = \tfrac{1}{\sqrt{2}}, \quad f_3(t) = \sin t. \quad \text{(Distinct!)}$$

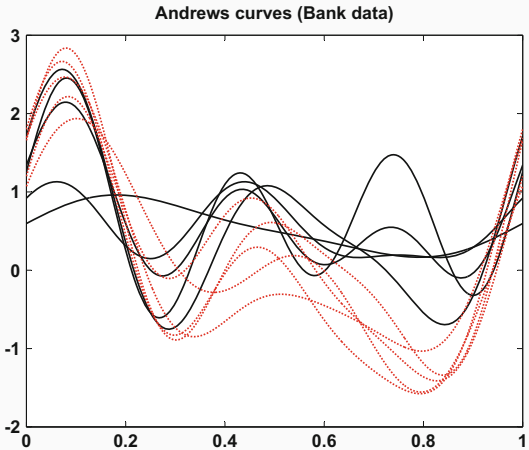
**Bank note example** (obs. 96, order $1, 2, \ldots, 6$):

$$f_{96}(t) = \tfrac{215.6}{\sqrt{2}} + 129.9\sin t + 129.9\cos t + 9.0\sin 2t + 9.5\cos 2t + 141.7\sin 3t.$$

# Andrews' Curves: Variable Order Matters




Order: $1, 2, 3, 4, 5, 6$




Order: $6, 5, 4, 3, 2, 1$

- **Black:** genuine (obs. 96–100); **Red dotted:** counterfeit (obs. 101–105).
- Original order: differences appear in **high frequency** (hard to see).

# Andrews' Curves Summary

| Summary: Andrews' Curves | |
|---|---|
| ↪ | Outliers appear as single curves **deviating from the bundle**. |
| ↪ | Sub-groups are characterised by **sets of similar curves**. |
| ↪ | The **order of variables** strongly affects interpretation. |
| ↪ | Optimal ordering is provided by **PCA** (Chapter 11). |
| ↪ | For $> 20$ observations: bad signal-to-ink ratio (Tufte, 1983). |

# Section 7: Parallel Coordinates Plots

# PCP: Concept and Construction

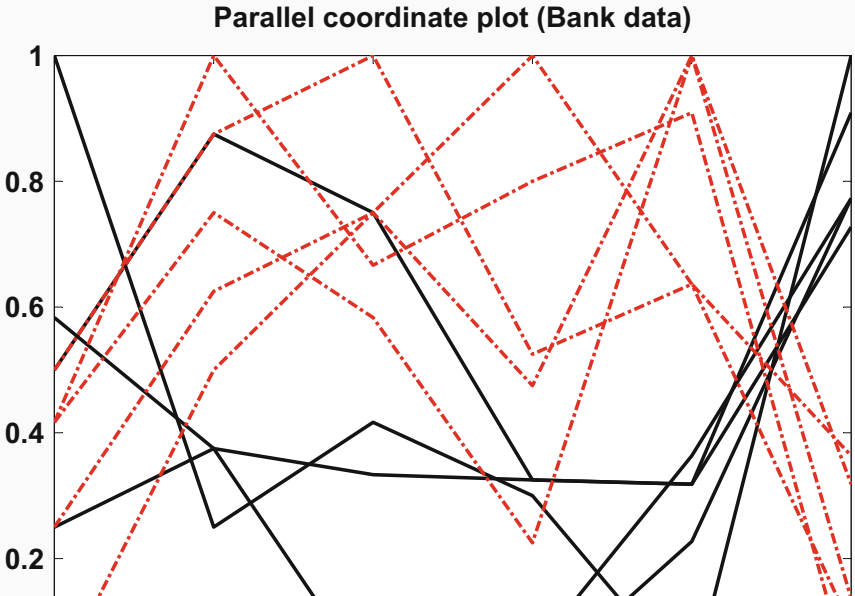
**Parallel Coordinates Plot (PCP)** (Inselberg, 1985):

- Draw $p$ parallel vertical axes (one per variable).
- Scale all variables to $[\min, \max] = [0, 1]$.
- Represent each observation $x_i$ as a **polyline** connecting its scaled values on the $p$ axes.

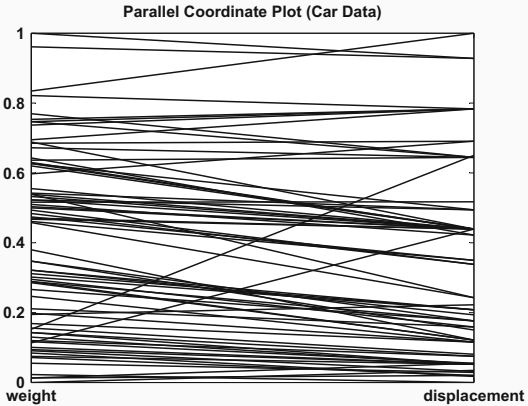
**Advantages over Cartesian plots:**

- Works for $p > 4$ (no dimensionality barrier).
- Positive dependence: lines approximately **parallel**.
- Negative dependence: lines **intersect** near the midpoint.
- Discrete variables and sub-groups are **visible**.

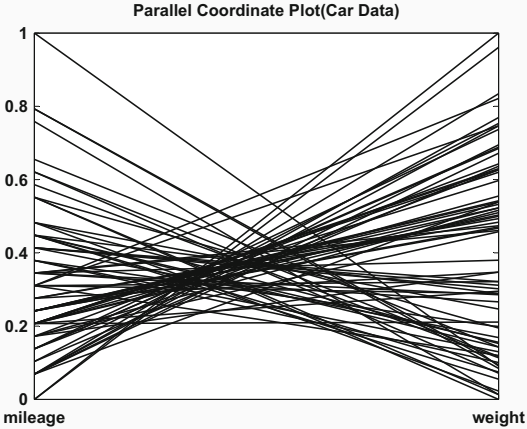
# PCP for Bank Note Observations 96–105



# Detecting Linear Dependence



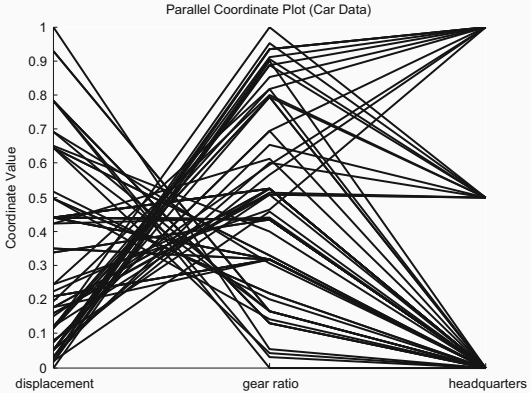

Weight vs Displacement ($\rho = 0.9$)



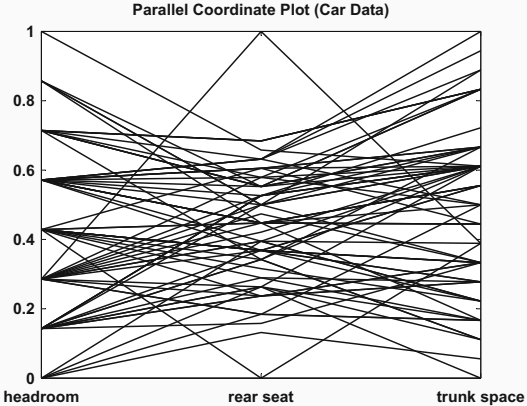

Mileage vs Weight ($\rho = -0.82$)

- **Near-parallel lines** $\Rightarrow$ strong **positive** correlation.
- **Lines crossing near midpoint** $\Rightarrow$ strong **negative** correlation.

# PCP for Sub-Group Detection and Outliers



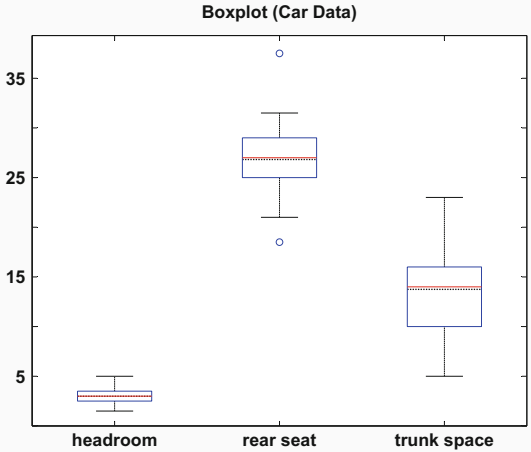

Displacement, Gear Ratio, Headquarters

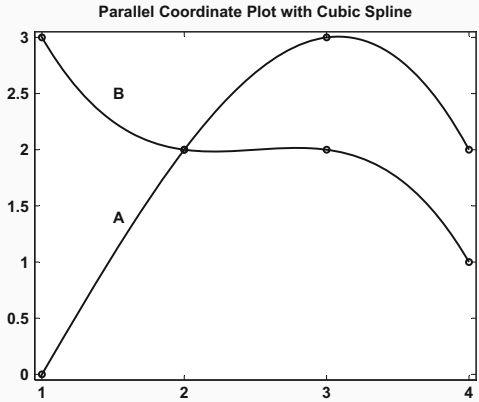



Headroom, Rear Seat, Trunk Space

- **Left:** Convergence to 3 discrete headquarters values (USA / Japan / EU).
- **Right:** Two **outliers** clearly visible — confirmed by the boxplot below.

# Boxplot Confirmation and Cubic Spline PCP

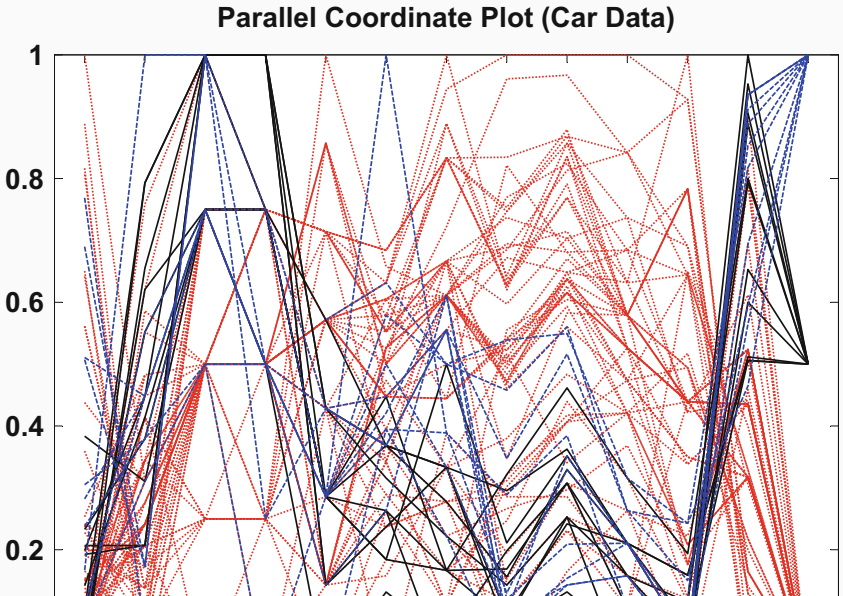



Cubic spline PCP

- **Crossing ambiguity** in standard PCP (obs. $A$ and $B$ sharing a value at $j = 2$): unresolvable with straight lines.
- **Cubic spline** interpolation resolves which curve belongs to which observation.

# Full Car Data PCP with Colouring



# PCP Summary

| Summary: Parallel Coordinates Plots | |
|---|---|
| ↪ | PCP overcomes the Cartesian limitation for dimensions $> 4$. |
| ↪ | Outliers appear as **outlying polygon curves**. |
| ↪ | Near-parallel lines indicate **positive** dependence; crossing lines indicate **negative** dependence. |
| ↪ | Variable ordering matters; sub-groups may be revealed by selective **colouring**. |
| ↪ | Cubic spline interpolation resolves crossing ambiguity. |

# Section 8: Hexagon Plots

# Motivation and Algorithm

**Problem:** For large $n$, ordinary scatterplots suffer from **overplotting** (many points on top of each other).

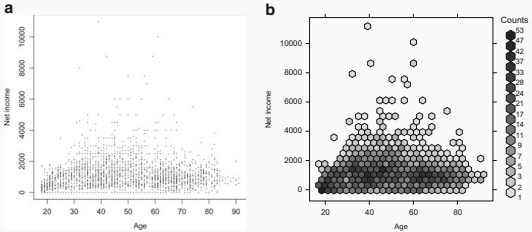
**Hexagon binning** (Lewin-Koh, 2006):

1. Tessellate the $(x, y)$-plane with a regular grid of hexagons.
2. Count the number of points $n_k$ in each hexagon $k$.
3. Colour or scale hexagons in proportion to $n_k$.

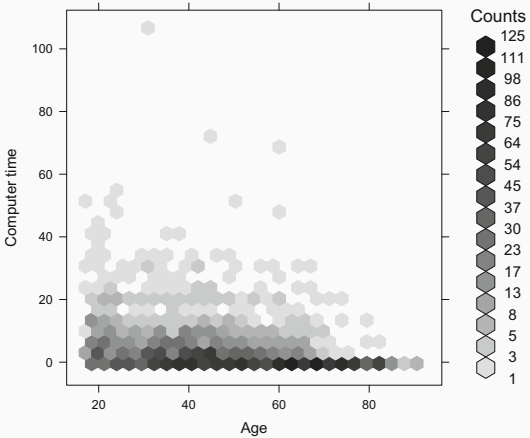
**Advantages of hexagons over squares:**

- **Symmetry of nearest neighbours** (equidistant centres).
- Maximum number of sides for a regular plane tessellation.
- Less visually biased for displaying densities.
- Extremely fast — works for $n \geq 10^6$.

# Hexagon Plots: Age vs Income and Age vs Computer Time




$X_1$ (Age) vs $X_2$ (Net income)




$X_1$ (Age) vs $X_5$ (Computer time)

ALLBUS 2006 survey data ($n = 2{,}946$). Most respondents earn $< 2{,}000$ EUR/month; $\approx 40\%$ of all age groups use a computer $< 1$ h/week.

# Hexagon Plot: Income vs Flat Size

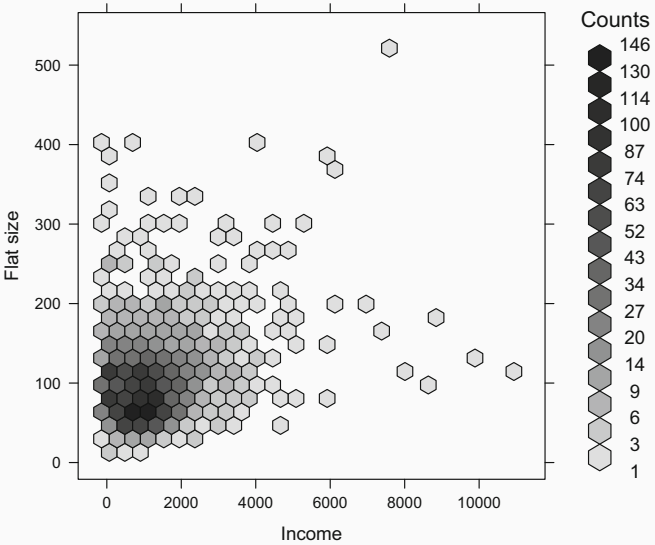



Figure 10: Hexagon plot: $X_2$ (Net income) vs $X_7$ (Flat size).

# Hexagon Plot Summary

| Summary: Hexagon Plots | |
|---|---|
| ↪ | Hexagon binning is a **bivariate histogram** for large data. |
| ↪ | Variance and bias trade off in opposite directions with bin width. |
| ↪ | Hexagons have the property of **symmetry of nearest neighbours**. |
| ↪ | Hexagons are visually **less biased** for density display than squares. |

# Section 9: Boston Housing

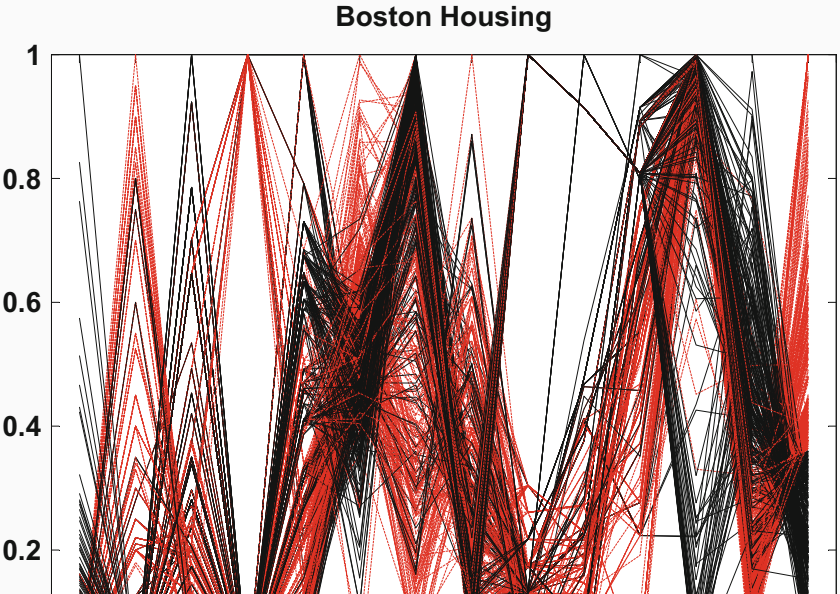
# Dataset Overview

**Boston Housing** (Harrison & Rubinfeld, 1978):

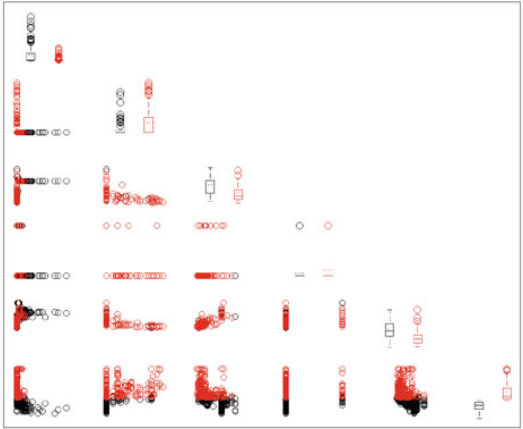
- **Goal:** Does "clean air" influence house prices?
- $n = 506$ census tracts in the Boston metropolitan area.
- 14 variables $X_1, \dots, X_{14}$ (see Chapter 22 for full description).
- $X_{14}$: median value of owner-occupied homes (target).

**Key variables:** $X_1$ crime rate, $X_3$ non-retail business, $X_5$ nitric oxide concentration, $X_6$ rooms/dwelling, $X_{11}$ pupil/teacher ratio, $X_{13}$ lower-status population, $X_{14}$ median home value.

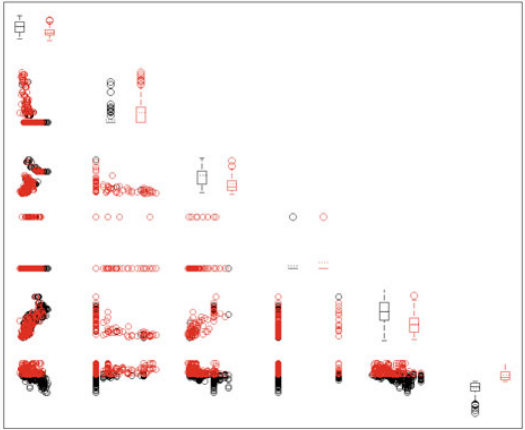
# PCP for Boston Housing



# Scatterplot Matrices (Original and Transformed)



$X_1, \ldots, X_5, X_{14}$



$\tilde{X}_1, \ldots, \tilde{X}_5, \tilde{X}_{14}$

- **Right:** Transformed data show more linear relationships.
- Black = cheaper districts; red = expensive districts.

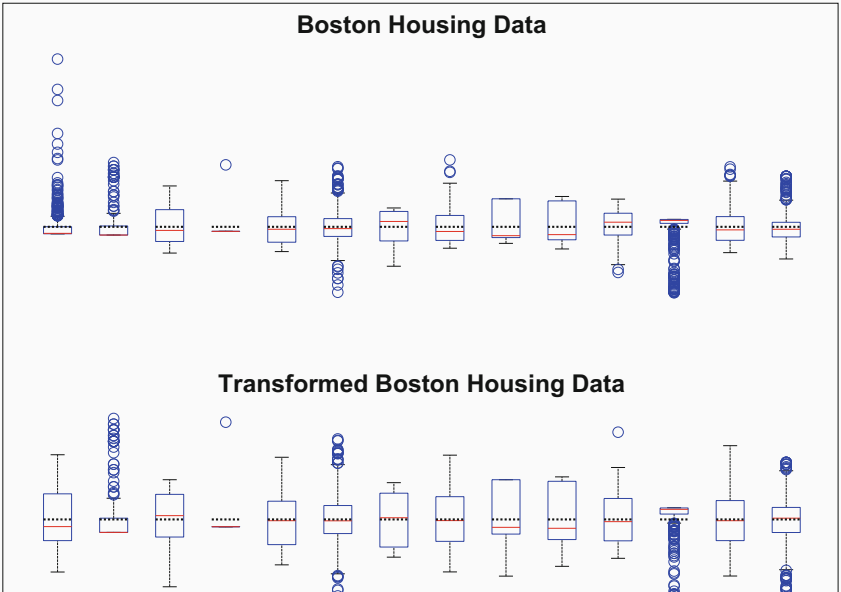
# Proposed Transformations

To reduce asymmetry in the marginal distributions:

$$\tilde{X}_1 = \log X_1, \qquad \tilde{X}_2 = X_2/10, \qquad \tilde{X}_3 = \log X_3,$$
$$\tilde{X}_5 = \log X_5, \qquad \tilde{X}_6 = \log X_6, \qquad \tilde{X}_7 = X_7^{2.5}/10000,$$
$$\tilde{X}_8 = \log X_8, \qquad \tilde{X}_9 = \log X_9, \qquad \tilde{X}_{10} = \log X_{10},$$
$$\tilde{X}_{11} = e^{0.4X_{11}}/1000, \qquad \tilde{X}_{12} = X_{12}/100, \qquad \tilde{X}_{13} = \sqrt{X_{13}},$$
$$\tilde{X}_{14} = \log X_{14}.$$

Note: $X_4$ is binary — **no transformation**.

# Boxplots Before and After Transformation



# Key Variable Relationships

| Variable | Relation to $X_{14}$ | Remark |
|---|---|---|
| $X_1$ (crime) | Negative | Two sub-groups visible in $\log X_1$ |
| $X_3$ (non-retail) | Negative | Bimodal KDE |
| $X_5$ (NOx) | Negative | May proxy noise pollution |
| $X_6$ (rooms) | **Positive** | Strongest linear signal |
| $X_{11}$ (pupil/teacher) | Negative | Fewer teachers $\Rightarrow$ lower prices |
| $X_{13}$ (lower status) | **Strongest negative** | Nearly linear on log scale |

# Exercises

# Selected Exercises

**Exercise 1.3** Assume $X \sim N(0,1)$. What percentage of data falls outside the outside bars $[F_L - 1.5d_F,\ F_U + 1.5d_F]$?

**Exercise 1.9** Use a bandwidth selection criterion to find the optimal $h$ for the diagonal variable of the bank notes. Is one bandwidth for both groups better?

**Exercise 1.10** The two density groups overlap near diagonal $\approx 140.4$. Can we effectively separate them using $X_6$ alone?

**Exercise 1.15** Heights of 13 students:

$$1.72,\ 1.83,\ 1.74,\ 1.79,\ 1.94,\ 1.81,\ 1.66,\ 1.60,\ 1.78,\ 1.77,\ 1.85,\ 1.70,\ 1.76.$$

(1) Five-number summary. (2) Boxplot. (3) Histogram.

**Exercise 1.18** Draftman plot for car data ($X_1 =$ price, $X_2 =$ mileage, $X_8 =$ weight, $X_9 =$ length). Brush on heavy cars. What can you say about price, mileage, length?

# Chapter Summary

Methods covered in Chapter 1:

| Method | Detects | Limitation |
|---|---|---|
| Boxplot | Location, spread, outliers, skew | No multimodality |
| Histogram | Modes, distribution shape | Sensitive to $h$ and $x_0$ |
| KDE | Modes, shape (smooth) | Bandwidth choice |
| Scatterplot | Bivariate/trivariate dependence | Max 3D |
| Faces | Cluster structure, outliers | Variable assignment |
| Andrews' curves | Clusters, outliers (curves) | Variable order |
| PCP | High-dim. structure, clusters | Crossing ambiguity |
| Hexagon plot | Large data density | Bin width choice |

**Recurring dataset:** Swiss bank notes — genuine vs. counterfeit ($n = 200, p = 6$).